Chapter 5: Laws of Motion

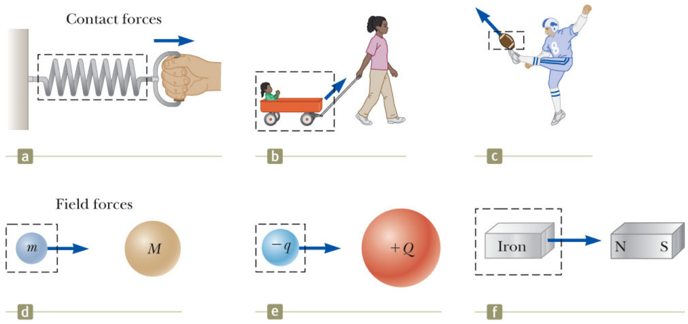
Concept of Force
Inertial Frames
Gravitational Force and Weight
Analysis using N2

Newton's 1st Law
Newton's 2nd Law
Newton's 3rd Law
Friction

The Laws of Motion

- Kinematics includes an objects position, velocity, and acceleration.
- Not what might influence that motion.
- Two main factors influencing the change of motion:
  - Forces acting on the object
  - The mass of the object
- Dynamics
- Will start with Newton's three laws of motion

Classes of Forces

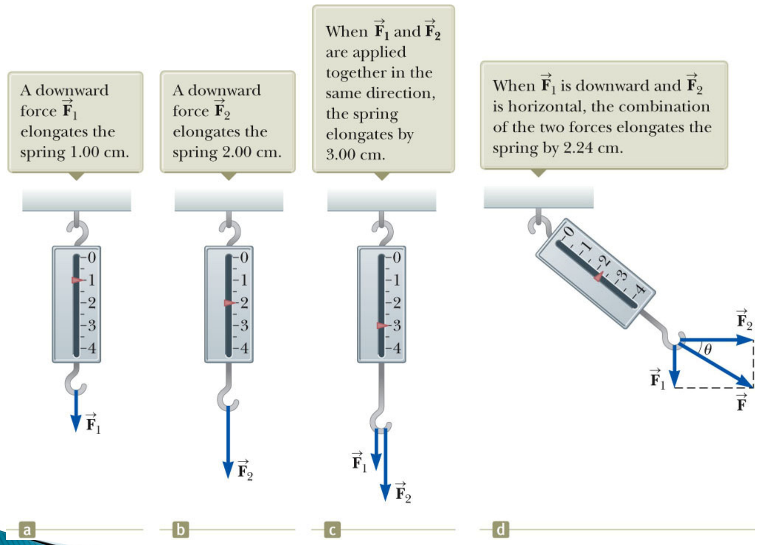


- Contact forces involve physical contact between two objects
- Field forces act through empty space
  - No physical contact is required

Fundamental Forces

- Gravitational force
  - Between objects
- Electromagnetic forces
  - Between electric charges
- Nuclear force (strong forces)
  - Between subatomic particles
- Weak forces
  - Arise in certain radioactive decay processes
- Note: These are all field forces.

More About Forces



Newton's First Law

- If an object does not interact with other objects, it is possible to identify a reference frame in which the object has zero acceleration.
  - This is also called the *law of inertia.*
  - It defines a special set of reference frames called *inertial frames.*
- Any reference frame that moves with constant velocity relative to an inertial frame is itself an inertial frame.
- If you accelerate relative to an object in an inertial frame, you are observing the object from a **non-inertial reference frame.**

Newton's First Law – Alternative Statement

- In the absence of external forces, when viewed from an inertial reference frame, an object at rest remains at rest and an object in motion continues in motion with a constant velocity.
  - Newton's First Law describes what happens in the absence of a force.
    - Does not describe zero net force
  - Also tells us that when no force acts on an object, the acceleration of the object is zero
  - Can conclude that any isolated object is either at rest or moving at a constant velocity
- The First Law also allows the definition of ***force*** as ***that which causes a change in the motion of an object.***

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Inertia and Mass

- The tendency of an object to resist any attempt to change its velocity is called ***inertia.***
- ***Mass*** is that property of an object that specifies how much resistance an object exhibits to changes in its velocity.
- $m_1/m_2 \equiv a_2/a_1$
- Weight is equal to the magnitude of the gravitational force exerted on the object.
  - Weight will vary with location (on earth and elsewhere).
  - w_{earth} = 784 N; w_{moon} ~ 130N
  - m_{earth} = 80 kg; m_{moon} = 80 kg

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Newton's Second Law

- When viewed from an inertial reference frame, the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass.
  - Force is the cause of *changes* in motion, as measured by the acceleration.
    - Remember, an object can have motion in the absence of forces.
- Algebraically,

$$\vec{a} \propto \frac{\sum \vec{F}}{m}$$

$$\sum \vec{F} = m\vec{a}$$

  - With a proportionality constant of 1 and speeds much lower than the speed of light.

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More About Newton's Second Law

- $\sum \vec{F}$ is the net force
  - This is the vector sum of all the forces acting on the object.
- Newton's Second Law can be expressed in terms of components:
  - $\Sigma F_x = ma_x$
  - $\Sigma F_y = ma_y$
  - $\Sigma F_z = ma_z$
- Remember that ma is not a force.
  - The sum of the forces is equated to this product of the mass of the object and its acceleration.
- The SI unit of force is the **newton** (N).
  - 1 N = 1 kg·m / s^2

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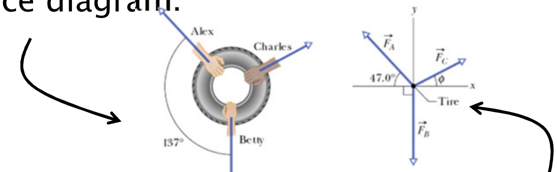
Newton's 2nd Law

- Equilibrium –there is no net force.
- Force diagram: only one object is shown, and all the forces acting on this object are depicted on the force diagram.



- Free body diagram: object is a point and the forces are drawn from the origin of the co-ordinate system.
- External force: in a system of 1 or more bodies, any force from outside the system on the bodies is an external force.
- Internal force: we do not include these in the FB diag.

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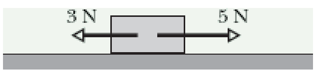
Example

- The figure here shows two horizontal forces acting on a block on a frictionless floor. If a third horizontal force $\vec{F}_3$ also acts on the block, what are the magnitude and direction of $\vec{F}_3$ when the block is (a) stationary and (b) moving to the left with a constant speed of 5 m/s?



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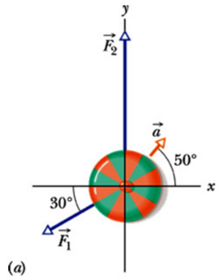
Example

In the overhead view of a 2.0 kg box is accelerated at 3.0 m/s^2 in the direction shown by $\vec{a}$ over a frictionless horizontal surface. The acceleration is caused by three horizontal forces, only two of which are shown: $\vec{F}_1$ of magnitude 10 N and $\vec{F}_2$ of magnitude 20 N. What is the third force $\vec{F}_3$ in unit-vector notation and in magnitude-angle notation?



Example

Gravitational Force (also ch. 13)

- The gravitational force, $\vec{F}_g$, is the force that the earth exerts on an object.
- This force is directed toward the center of the earth.
- From Newton's Second Law:
  - $\vec{F}_g = G\dfrac{mM_e}{r^2}\hat{r} = m\vec{a}$

    $a = G\dfrac{M_e}{r^2} = g$

    $F_g = mg = W$
- G = 6.674×10^{-11} N·m^2/kg^2

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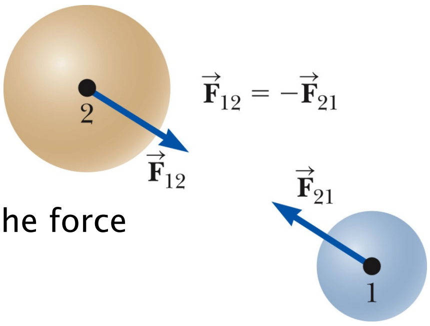
Newton's Third Law

- If two objects interact, the force $\vec{F}_{12}$ exerted by object 1 on object 2 is equal in magnitude and opposite in direction to the force $\vec{F}_{21}$ exerted by object 2 on object 1.
  - $\vec{F}_{12} = -\vec{F}_{21}$



- Note on notation: $\vec{F}_{12}$ is the force exerted *by* 1 *on* 2.

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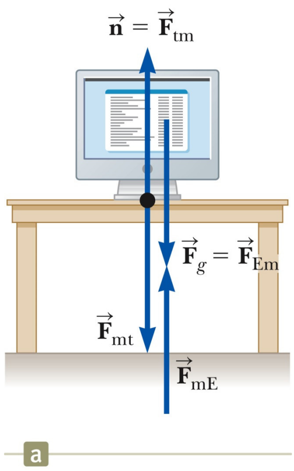
Newton's Third Law, Alternative Statement

- The action force is equal in magnitude to the reaction force and opposite in direction.
  - One of the forces is the action force, the other is the reaction force.
  - It doesn't matter which is considered the action and which the reaction.
  - The action and reaction forces must act on different objects and be of the same type.

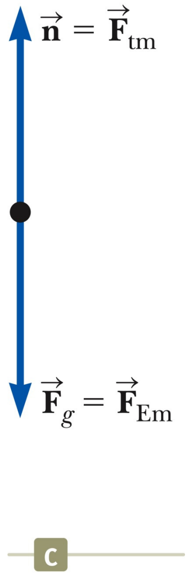
Action-Reaction Examples

- The normal force (table on monitor) is the reaction of the force the monitor exerts on the table.
  - Normal means perpendicular, in this case
- The action (Earth on monitor) force is equal in magnitude and opposite in direction to the reaction force, the force the monitor exerts on the Earth.



Free Body Diagram

- The most important step in solving problems involving Newton's Laws is to draw the free body diagram.
- Be sure to include only the forces acting on the object of interest.
- Include any field forces acting on the object.
- Do not assume the normal force equals the weight.



Free Body Diagrams and the Particle Model

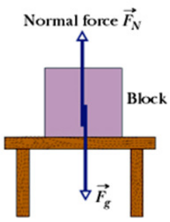
- The particle model is used by representing the object as a dot in the free body diagram.
- The forces that act on the object are shown as being applied to the dot.
- The free body helps isolate only those forces acting on the object and eliminate the other forces from the analysis.

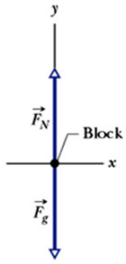
Normal Force

- Normal Force: When an object presses against a surface, the surface deforms and pushes on the body with a normal force perpendicular to the contact surface.
- Symbol is: $\vec{F}_N$ or $\vec{N}$ or $\vec{n}$
- It is always perpendicular to the surface on which the object lies.
- Example – Block on a table has only 2 forces acting on it – no acceleration
- In this case: $F_{net,y} = F_N - F_g = ma_y$

$$F_N = mg + ma_y = m(g + a_y)$$

$$F_N = mg$$





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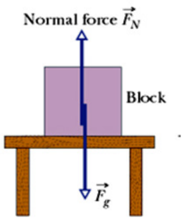
Example

Is the magnitude of the normal force greater, less than or equal to mg if the block and table are in a lift moving upward
(a) at constant speed?
(b) at increasing speed?



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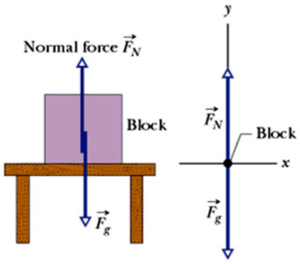
Example

Is the magnitude of the normal force greater, less than or equal to mg if the block and table are in a lift moving upward
(a) at constant speed?
(b) at increasing speed?

$$\Sigma F_y = ma_y = F_N - mg$$
$$F_N = m\left(g + a_y\right)$$

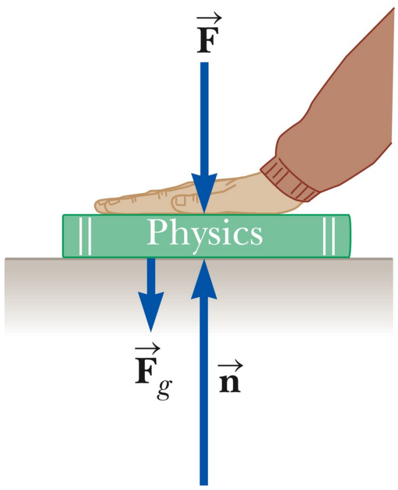


(a) Equal to mg, since $a_y = 0$

(b) Greater than mg, since $a_y > 0$

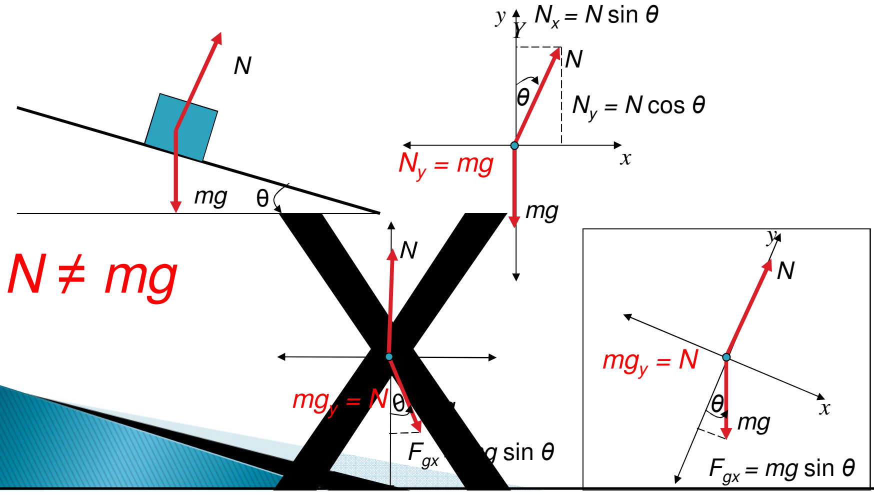
Example

- For example, in this case
  $\sum F_y = n - F_g - F = 0$
  and $n = mg + F$
- $\vec{n}$ may also be less than $\vec{F}_g$



Normal Force

- Normal force – stationary object on a slope



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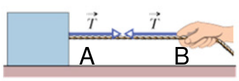
Tension

- Tension: This is the force exerted by a rope or a cable attached to an object
- Tension has the following characteristics:
  - It is always directed along the rope.
  - It is always pulling the object.
  - It has the same value along the rope (for example, points A and B).



- The following assumptions are made:
  - The rope has NEGLIBIBLE MASS compared to the mass of the object it pulls.
  - The rope does NOT STRETCH.

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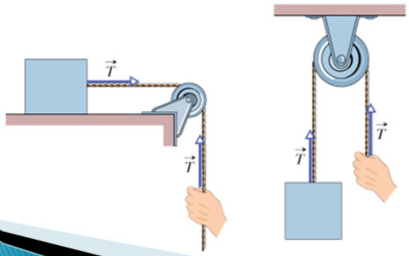
Tension (2)

- Tension: This is the force exerted by a rope or a cable attached to an object
- If a pulley is used as in the figure, we assume that the pulley is massless and frictionless (in the following 2 chapters.
- Remember the tension is the same along the length of the rope.



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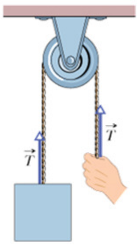
Example

- The suspended body in the figure weighs 75 N. Is T equal to, greater than or less than 75 N when the body is moving upward
  (a) at constant speed,
  (b) at increasing speed, and
  (c) at decreasing speed?



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Analysis Models Using Newton's Second Law

- Assumptions
  - Objects can be modeled as particles.
  - Interested only in the external forces acting on the object
    - Can neglect internal action-reaction forces
  - Initially dealing with frictionless surfaces
  - Masses of strings or ropes are negligible.
    - The force the rope exerts is away from the object and parallel to the rope.
    - When a rope attached to an object is pulling it, the magnitude of that force is the **tension** in the rope.

Analysis Model: The Particle in Equilibrium

- If the acceleration of an object that can be modeled as a particle is zero, the object is said to be in **equilibrium.**
  - The model is the *particle in equilibrium model.*
- Mathematically, the net force acting on the object is zero.

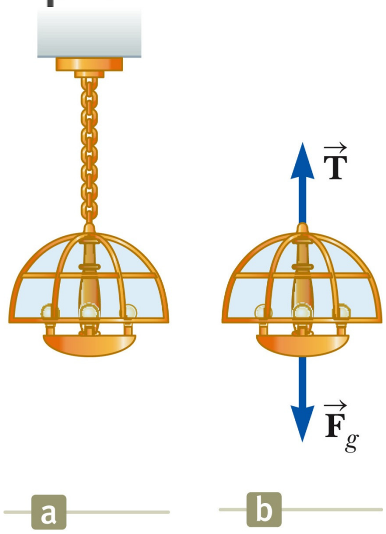
$$\sum \vec{\mathbf{F}} = 0$$

$$\sum F_x = 0 \text{ and } \sum F_y = 0$$

Equilibrium, Example

- A lamp is suspended from a chain of negligible mass.
- The forces acting on the lamp are:
  - the downward force of gravity
  - the upward tension in the chain
- Applying equilibrium gives

$$\sum F_y = 0 \rightarrow T - F_g = 0 \rightarrow T = F_g$$



Analysis Model: The Particle Under a Net Force

- If an object that can be modeled as a particle experiences an acceleration, there must be a non-zero net force acting on it.
  - Model is *particle under a net force model.*
- Draw a free-body diagram.
- Apply Newton's Second Law in component form.

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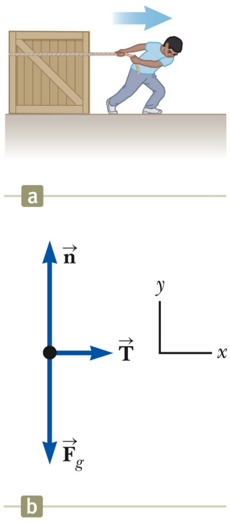
Newton's Second Law, Example 1

- Forces acting on the crate:
  - A tension, acting through the rope, is the magnitude of force $\vec{T}$
  - The gravitational force, $\vec{F}_g$
  - The normal force, $\vec{n}$, exerted by the floor



Newton's Second Law, Example 1

cont.

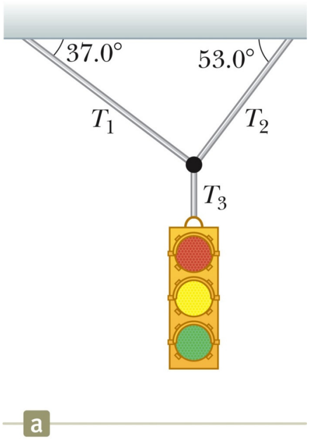
- Apply Newton's Second Law in component form:

$$\sum F_x = T = ma_x$$
$$\sum F_y = n - F_g = 0$$
$$n = F_g$$

- Solve for the unknown(s)
- If the tension is constant, then *a* is constant and the kinematic equations can be used to more fully describe the motion of the crate.

Example

- Example 5.4
- A traffic light weighing 122 N hangs from a cable fastened to a support as in the Figure. The upper cables make angles of $\theta_1 = 37.0°$ and $\theta_2 = 53.0°$ with the horizontal. These upper cables are not as strong as the vertical cable, and will break if the tension in them exceeds 100 N. Does the traffic light remain hanging in this situation, or will one of the cables break?



Problem-Solving Hints – Applying Newton's Laws

- *Conceptualize*
  - Draw a diagram
  - Choose a convenient coordinate system for each object
- *Categorize*
  - Is the model a particle in equilibrium?
    - If so, $\Sigma F = 0$
  - Is the model a particle under a net force?
    - If so, $\Sigma F = m\,a$

Problem-Solving Hints – Applying Newton's Laws, cont.

- Analyze
  - Draw free-body diagrams for each object
  - Include only forces acting on the object
  - Find components along the coordinate axes
  - Be sure units are consistent
  - Apply the appropriate equation(s) in component form
  - Solve for the unknown(s)
- Finalize
  - Check your results for consistency with your free-body diagram
  - Check extreme values

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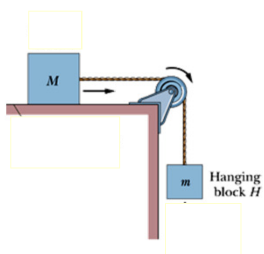
Example

- The Figure shows a block S (the sliding block) with mass $M = 3.3$ kg. The block is free to move along a horizontal frictionless surface and connected, by a cord that wraps over a frictionless pulley, to a second block H (the hanging block), with mass $m = 2.1$ kg. The cord and pulley have negligible masses compared to the blocks (they are "massless"). The hanging block H falls as the sliding block S accelerates to the right. Find (a) the acceleration of block S, (b) the acceleration of block H, and (c) the tension in the cord.



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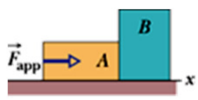
Example

- $M = 3.3$ kg $m = 2.1$ kg
  Find (a) the acceleration of block S, (b) the acceleration of block H, and (c) the tension in the cord.
- Draw free-body diagram of each block

Example

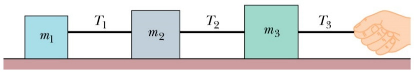
In the Figure, a constant horizontal force of magnitude 20 N is applied to block A of mass $m_A = 4.0$ kg, which pushes against block B of mass $m_B = 6.0$ kg. The blocks slide over a frictionless surface, along an x axis. (a) What is the acceleration of the blocks? (b) What is the (horizontal) force on block B from block A?



(a)

Example

In the figure, three connected blocks are pulled to the right on a horizontal frictionless table by a force of magnitude $T_3 = 65.0$ N. If $m_1 = 12$ kg, $m_2 = 24.0$ kg and $m_3 = 31.0$ kg, calculate (a) the magnitude of the system's acceleration, (b) the tension T_1 and (c) the tension T_2.



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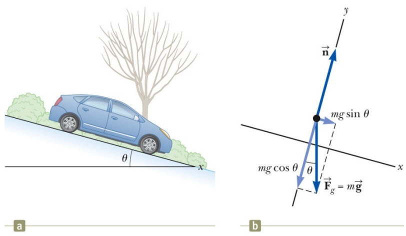
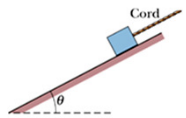
Inclined Planes

- Categorize as a particle under a net force since it accelerates.
- Forces acting on the object:
  - The normal force acts perpendicular to the plane.
  - The gravitational force acts straight down.
- Choose the coordinate system with x along the incline and y perpendicular to the incline.
- Replace the force of gravity with its components.
- Apply the model of a particle under a net force to the x-direction and a particle in equilibrium to the y-direction.



Example

- In the figure, a cord pulls on a box up along a frictionless plane inclined at θ = 30°. The box has mass m = 5.00 kg, and the force from the cord has magnitude T = 25.0 N. What is the box's acceleration component a along the inclined plane?



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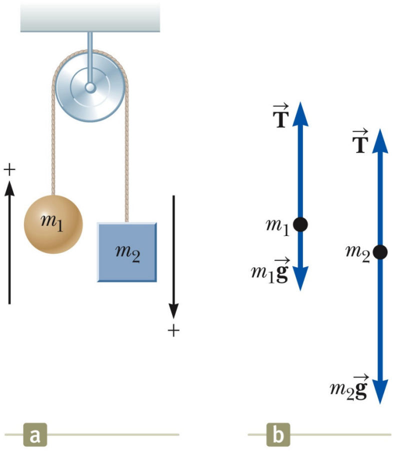
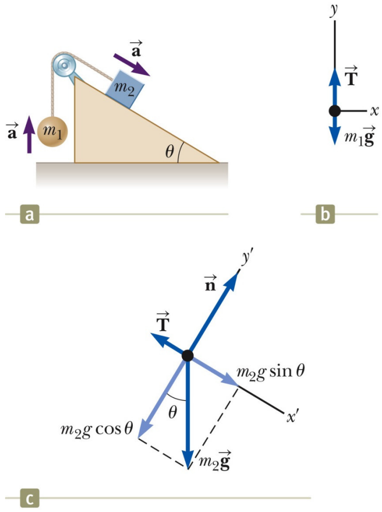
Multiple Objects, Example – Atwood's Machine

- Forces acting on the objects:
  - Tension (same for both objects, one string)
  - Gravitational force
- Each object has the same acceleration since they are connected.
- Draw the free-body diagrams
- Apply Newton's Laws
- Solve for the unknown(s)



Multiple Objects, Example 2

- Draw the free-body diagram for each object
  - One cord, so tension is the same for both objects
  - Connected, so magnitude of acceleration is the same for both objects
- Categorize as particles under a net force
- Apply Newton's Laws
- Solve for the unknown(s)



Forces of Friction

- When an object is in motion on a surface or through a viscous medium, there will be a resistance to the motion.
  - This is due to the interactions between the object and its environment.
- This resistance is called the *force of friction.*

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Forces of Friction, cont.

- Friction is proportional to the normal force.
  - $f_s \leq \mu_s\ n$ and $f_k = \mu_k\ n$
    - μ is the **coefficient of friction**
  - These equations relate the magnitudes of the forces; they are not vector equations.
  - For static friction, the equals sign is valid only at *impeding* motion, the surfaces are on the verge of slipping.
  - Use the inequality for static friction if the surfaces are not on the verge of slipping.

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Forces of Friction, final

- The coefficient of friction depends on the surfaces in contact.
- The force of static friction is generally greater than the force of kinetic friction.
- The direction of the frictional force is opposite the direction of motion and parallel to the surfaces in contact.
- The coefficients of friction are nearly independent of the area of contact.

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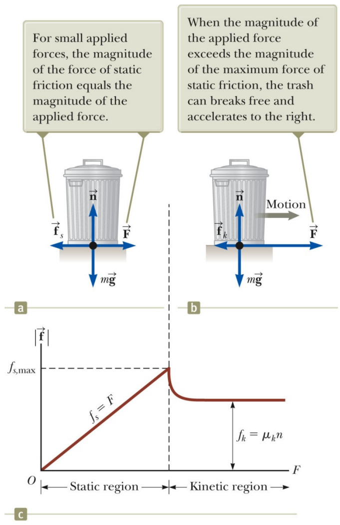
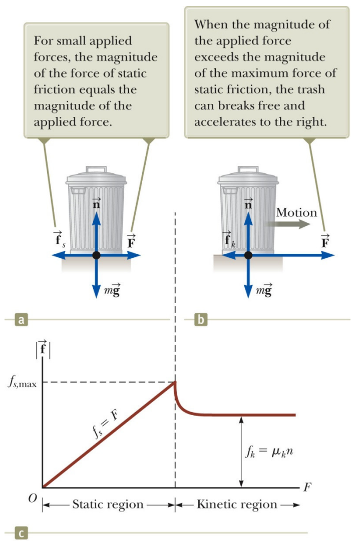
Static Friction

- Static friction acts to keep the object from moving.
- As long as the object is not moving, $f_s = F$
- If $\vec{\mathbf{F}}$ increases, so does $\vec{f}_s$
- If $\vec{\mathbf{F}}$ decreases, so does $\vec{f}_s$
- $f_s \le \mu_s n$
  - Remember, the equality holds when the surfaces are on the verge of slipping.



Explore Forces of Friction

- Vary the applied force
- Note the value of the frictional force
  - Compare the values
- Note what happens when the can starts to move



Section 5.8

Some Coefficients of Friction

TABLE 5.1 *Coefficients of Friction*

	μ_s	μ_k
Rubber on concrete	1.0	0.8
Steel on steel	0.74	0.57
Aluminum on steel	0.61	0.47
Glass on glass	0.94	0.4
Copper on steel	0.53	0.36
Wood on wood	0.25–0.5	0.2
Waxed wood on wet snow	0.14	0.1
Waxed wood on dry snow	—	0.04
Metal on metal (lubricated)	0.15	0.06
Teflon on Teflon	0.04	0.04
Ice on ice	0.1	0.03
Synovial joints in humans	0.01	0.003

Note: All values are approximate. In some cases, the coefficient of friction can exceed 1.0.

Friction in Newton's Laws Problems

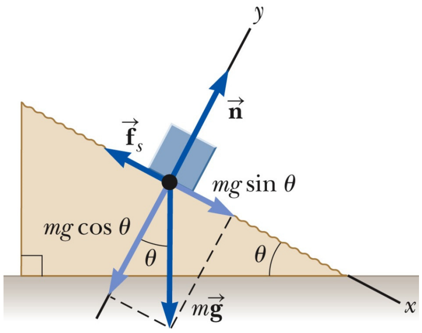
- Friction is a force, so it simply is included in the $\sum \vec{\mathbf{F}}$ in Newton's Laws.
- The rules of friction allow you to determine the direction and magnitude of the force of friction.

Section 5.8

Friction Example, 1

Suppose a block is placed on a rough surface inclined relative to the horizontal as shown in the Figure. The incline angle is increased until the block starts to move. Show that you can obtain μ_s by measuring the critical angle θ_c at which this slipping just occurs.

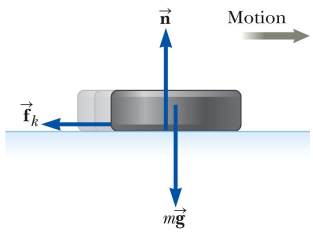
- $\mu = \tan\theta$
  - For μ_s, use the angle where the block just slips.
  - For μ_k, use the angle where the block slides down at a constant speed.



Friction, Example 2

A hockey puck on a frozen pond is given an initial speed of 20.0 m/s. If the puck always remains on the ice and slides 115 m before coming to rest, determine the coefficient of kinetic friction between the puck and the ice.

- Draw FB Diagram
- Continue with the solution as with any Newton's Law problem.
- What is the direction of the friction?



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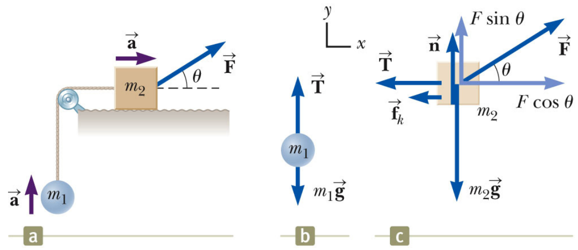
Friction, Example 3



- A block of mass m_2 on a rough, horizontal surface is connected to a ball of mass m_1 by a lightweight cord over a lightweight pulley. A force of magnitude F at an angle θ with the horizontal is applied to the block as shown and the block slides to the right. The coefficient of kinetic friction between the block and the surface is μ_s . Determine the magnitude of the acceleration of the two objects.
- Draw the free-body diagrams.
- Apply Newton's Laws as in any other multiple object system problem.

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Analysis Model Summary

- Particle under a net force
  - If a particle experiences a non-zero net force, its acceleration is related to the force by Newton's Second Law.
  - May also include using a particle under constant acceleration model to relate force and kinematic information.
- Particle in equilibrium
  - If a particle maintains a constant velocity (including a value of zero), the forces on the particle balance and Newton's Second Law becomes:

$$\sum \vec{F} = 0$$

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